

Final (practice version - NOT GRADED), Math 170a, Fall 2018
Instructor: Elizaveta Rebrova

Printed name: _____

Signed name: _____

Student ID number: _____

Instructions:

- Read problems very carefully. If you have any questions please ask.
- The correct final answer alone is not sufficient for full credit - you should explain your answers as much as you can, saving enough time to attempt all the problems.
- Your final answers do not need to be completely simplified (unless otherwise stated), e.g. a sum of several decimal fractions would be completely fine. Ask if in doubt.
- You are allowed to use only the items necessary for writing. Notes, books, sheets, calculators, phones are not allowed, please put them away. If you need more paper, please raise your hand.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
8	10	
9	10	
10	10	
Total:	100	

1. For this problem only: please provide short answers (or short explanations when appropriate).
 - (a) (2 points) Let X be a Binomial random variable such that $\mathbb{E}(X) = 4$ and $\text{var}(X) = 2$. Find the parameters of X .
 - (b) (2 points) Are there random variables X and Y (not necessarily independent) both of which are uniform in $[0, 1]$ and such that $\mathbb{P}(X + Y = 1) > 0$?
 - (c) (2 points) Let $F_X(t)$ be the cumulative distribution function (CDF) of a continuous random variable X and let $Y = -X$. Express the CDF of Y terms of $F_X(t)$.
 - (d) (2 points) Let X be a standard normal random variable. Write down one function $f(t)$ so that the random variable $Y = f(X)$ is normal with mean a and variance b .
 - (e) (2 points) Assume that random variables X and Y defined on the same probability space satisfy $\mathbb{P}(X + Y \leq 0) = \mathbb{P}(X + Y \leq 2)$. Find the probability $\mathbb{P}(0 < X \leq 1, 0 < Y \leq 1)$.

2. (10 points) There are n boxes, the 1st one containing 1 glass, the 2nd box containing 2 glasses, the 3rd box containing 3 glasses, ... , k th box containing k glasses, ... n th box containing n glasses. In each box every glass is broken with probability $1/2$ independently of all other glasses. A contestant chooses a box uniformly at random. Given that they found no broken glasses in this box, find the conditional probability that they choose the 1st box. Simplify your answer.

3. (10 points) There are n participants in a psychological experiment. Participants are put in one room and are shown a red toy. After that, the researcher asks participants one by one (starting from the 1st participant till the n -th participant) about the color of the toy. Each participant is instructed to reply red or blue with equal probabilities. However, it is known that if k people have replied blue so far, the next participant will reply red only with probability $\frac{1}{2+k}$ (for each $k \geq 1$). For each $1 \leq l \leq n$ find the probability that the l -th participant is the first one who answered red. What is the probability that all participants answered blue? Simplify your answers.

4. (10 points) You arrive at the opening of a new cafe which has only two chairs. Unfortunately, there are three people ahead of you in the line. You know that the number of minutes that a guest spends in a cafe is a geometric random variable with mean 2, and that the time each guest spends in the cafe is independent of all other guests. What is the probability you will have to wait more than n minutes.

5. (10 points) A frog is sitting on a small lily pad at a uniform location in a square shaped pool of a side length one meter. Let X be the minimal distance the frog has to jump to escape the pool. Find the cumulative distribution function (CDF), probability density function (PDF) and the expectation of X .

6. (10 points) Let T be a triangle with vertices $(-1, 0)$, $(1, 0)$ and $(0, 1)$. Let the random variables X and Y be jointly continuous with joint PDF

$$f_{X,Y}(x, y) = \begin{cases} C|xy|, & \text{for } (x, y) \in T, \\ 0, & \text{for } (x, y) \notin T. \end{cases}$$

Find the value of the constant C , and the marginal PMFs for both X and Y . Are X and Y independent? Find the CDF of the random variable $Z = X + Y$ (compute the integral, but there is no need to simplify the resulting expression).

7. (10 points) Let X be an exponential random variable with parameter 1. Given $X = t$, let Y be uniformly distributed on the interval $[0, e^t]$. Compute $\mathbb{E}(Y|X = 1)$. Compute the joint PDF of X and Y and the marginal PDF of Y .

8. In a certain game a computer prints a random positive integer number N whose distribution is Poisson with parameter λ . Then another computer selects a real number X uniformly in the interval $[0, N + 1]$. Simplify the answers.
- (a) (5 points) Compute the PDF of X at all values between 0 and 1, that is compute $f_X(x)$ for all $0 < x < 1$.
 - (b) (5 points) Compute the expectation of X .

9. A box contains $2n$ red and $2n$ blue toys. We select uniformly at random $2n$ toys from the box.
- (a) (5 points) Compute the probability that we selected equally many red and blue toys.
 - (b) (5 points) Given that we selected more red than blue toys, compute the conditional probability that no blue toys were selected.

10. We have n fair coins at our disposal. We perform the experiment in which we toss these n coins independently. We discard the coins which came up tails, and keep those which came up heads. If we have no more coins (that is, if all coins came up tails) we end the experiment. Otherwise, we repeat the experiment with the coins which are left, so we toss the coins which are left and discard those which came up tails. We repeat this experiment until we have no more coins left. Let X denote the number of times we had to perform the experiment.
- (a) (5 points) Given that $X = 2$, compute the conditional probability that in the first experiment exactly one coin landed heads.
- (b) (5 points) Find the PMF of X .